Vridhi Brahmbhatt

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SUMMARY

Backend-focused full-stack engineer (Python/Django, TypeScript/Next.js) with a portfolio of production-style systems spanning settlement infrastructure, security tooling, and financial applications. Deep experience with async processing (Celery, Django Channels), graph algorithms, and AI-native development workflows. Currently seeking platform or backend engineering roles.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, Bash, Solidity
Backend:	Django, DRF, Celery, Redis, PostgreSQL, Django Channels, REST APIs, WebSockets
Frontend:	Next.js, React, TailwindCSS, TanStack Query, D3, Cytoscape
Algorithms:	Graph theory (Tarjan's SCC), priority-queue scheduling, rolling z-score, Isolation Forest
Tooling:	Git, GitHub Actions, Docker (basic), Linux (systemd, daphne), pytest, factory_boy
AI Workflow:	Claude, Cursor, GPT-4; disciplined review of AI output for use in scaffolding, refactoring, and test generation

EXPERIENCE

Independent Full-Stack Engineer
Self-Directed Projects 2023–Present

- Designed and shipped four production-style applications covering payment settlement, security monitoring, tokenized assets, and physical-world dispute ledgers.
- Built end-to-end services with Django REST Framework, Celery workers, Redis, PostgreSQL, and Django Channels for real-time WebSocket updates; established hermetic test suites (pytest, factory_boy) and GitHub Actions CI with in-memory channel layers.
- Implemented graph algorithms in production code, including an iterative Tarjan SCC for multilateral netting, and priority-queue settlement scheduling.
- Integrated AI workflows (Claude, Cursor) into daily development; shipped an anomaly detection feature using scikit-learn's Isolation Forest and rolling z-scores.

KEY PROJECTS

Obligation Net Optimizer | *Django, DRF, Celery, Redis, Channels, Next.js, D3, Cytoscape, scikit-learn*

- Architected a real-time multilateral netting and settlement engine that reduces gross payment obligations to minimal net payments via bilateral netting, iterative Tarjan SCC, and a priority-queue settlement scheduler.
- Built asynchronous pipeline with Celery + Redis; live dashboard updates pushed via Django Channels WebSocket to TanStack Query caches.
- Designed schema and persistence layer (Obligation, NettingWindow, NetPosition, SettlementAttempt, ParticipantBalance) with migrations, constraints, and ORM-level integrity.
- Added AI-driven anomaly detection (rolling z-score + Isolation Forest) with a dedicated UI for operator review and dismissal workflow.
- Full test suite (100+ tests) with hermetic CI via in-memory channel layers and eager Celery; zero external dependencies for CI runs.

LayerZero Infrastructure Security Scanner | *Django, DRF, PostgreSQL, Redis, Next.js, Discord API*

- Built a full-stack monitoring platform that scans LayerZero DVN/oracle/proxy configurations, scores risk via a custom engine, and delivers alerts via Discord webhooks.
- Designed REST APIs with DRF, JWT auth, Redis-based rate limiting, and public/private endpoint segregation.
- Created a Next.js dashboard with pagination, filtering, and search, while handling complex server state with TanStack Query.

Aurum Protocol: Tokenized Gold System | *Solidity, Foundry, Next.js, Chainlink*

- Built a full-stack Real World Asset (RWA) tokenization platform with a public dashboard showing live TVL, LTV ratios, and liquidation metrics.
- Designed a dynamic risk model (LTV, interest accrual, automatic debt allocation, liquidation curves) driven by Chainlink price feeds.
- Achieved 95% test coverage via unit, fuzz, and invariant tests; self-audited and fixed multiple critical business-logic edge cases.

EDUCATION & CERTIFICATIONS

B.S. Computer Science, Western Governor's University September 2025

- *Certifications:* ITIL Foundation in IT Service Management, Linux Professional Institute (LPI) Linux Essentials